

金子梁 JIN Ziliang

Assistant Professor

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Research interests

- Derivation and migration of primordial materials in the solar system
- Plasma-solid interactions in space environments
- Deep magmatic processes in the Moon's interior

Education

- 2014 – 2017 **Guest scientist** GeoForschungZentrum, Potsdam, Germany.
Co-Advisor: **Dr. Michael Wiedenbeck**
- 2012 – 2017 **Ph.D. candidate** China University of Geosciences, Beijing
Advisor: **Dr. Zhaochong Zhang**
- 2008 – 2012 **B.Sc. (Geology)** China University of Geosciences, Beijing

Professional appointment

- 2021 – present **Assistant Professor**–Macau University of Science and Technology
- 2017 – 2020 **Postdoctoral scholar**–Arizona State University, Tempe, U.S.
Supervisor: **Dr. Maitrayee Bose**

Professional experiences

Macau University of Science and Technology (2021-present)

- Manage LA-ICP-MS (GeoLasHD + Agilent 7900) and low-energy ion implanter
- Simulate hydration processes in early solar system materials via ion implantation
- Research on the Moon's redox evolution and early solar system material transport

Arizona State University, Arizona, U.S. (2017-2020)

- Maintained lab operations, purchasing, scheduling, and training
- Developed high-precision H measurement protocols using NanoSIMS 50L
- Studied hydrogen isotope systematics in meteorites and asteroids (SIMS, NanoSIMS 50L, IMS 6f)
- Developed Duoplasmatron and Cs ion sources (with Peter Williams)
- Contributed to building a Class 10,000 clean lab (PI: Maitrayee Bose)
- Operated NanoSIMS 50L on projects including:
 - Water/H isotopes in Martian meteorites (PI: Anne Peslier)

- Li diffusion in zircon (PI: Christy Till)
- Li isotopes in coal (PI: Lynda Williams)
- Presolar grain imaging (PI: Maitrayee Bose)
- C, O, H isotopes in carbonates (PI: Myriam Telus)
- C-N isotopes in bee brains (PI: Tyler Quigley)

GeoForschungsZentrum, Potsdam, Germany (2015-2017)

- Developed SIMS reference materials: synthetic silicate glass, olivine, and hornblende
- Studied SIMS sputtering behavior and isotopic homogeneity of O/C standards
- Conducted zircon U-Pb and oxygen isotope dating of tuff from E'meishan LIP (PI: Zhaochong Zhang)

Grants

- FDCT (2021-2023) **PI**: A comprehensive study of mineral disequilibrium in the achondrite EC002
- FDCT (2023-2024) **PI**: Process development: ion-implanted reference materials used for in-situ analysis
- NSFC (2023-2025) **PI**: The hydrogen implantation-diffusion process that hydrates the silicates in the solar nebula.
- FDCT (2025-2028) **Co-PI**: Research on key scientific and technological questions in extraterrestrial sample analysis.

Selected peer-reviewed articles

**Indicates corresponding author*

- P. Liu*, **Z. Jin***, Z. Gong, L. Li, K. Wang, X. Zeng, J. Liu, D. Song, H. Du*, K. Zhu, K. Yan, J. Li, K. Zhang, A. Hirt, W. Williams (2026). Magnetic vortex state of natural lunar γ -Fe. *Proceedings of the National Academy of Sciences (PNAS)*, in press.
- J. Xiang, W. Neumann, Y. Li, L. Jin, Y. Zhu, **Z. Jin*** (2026), Sorting and recycling processes of early inner Solar System materials. *Astronomy & Astrophysics*, 710, p.A324.
- X. Zeng, **Z. Jin***, M. Bose, R. Hervig, T. Long (2025), Effects of Crystal Orientation and Temperature on Space Irradiation of Silicates. *The Astrophysical Journal*, 991 (2), 159
- X. Zeng, **Z. Jin***, C. Dong, Z. Huang, M-H Zhu, L. Xu, L. Morrissey, L. Wang (2025), Earth Wind-Driven Formation of Hematite on the Lunar Surface. *Geophysical Research Letters*, 52 (17), e2025GL116170.

- **Z. Jin***, T. Hou, M. Zhu, Y. Zhang and O. Namur, (2025). Late-stage microstructures in Chang'E-5 basalt and implications for the evolution of lunar ferrobasalt. *American Mineralogist*, 110(4), pp.560-569.
- **Z. Jin***, Y. Zhang, M. Bose, S. Glynn, and F. Couffignal, (2024). Petrogenesis of Erg Chech 002 Achondrite and Implications for an Altered Magma Ocean. *The Astrophysical Journal*, 965(1), p.24.
- **Z. Jin***, M. Bose, T. Lichtenberg, G. Mulders (2021), New evidence for wet accretion of inner solar system planetesimals from meteorites Chelyabinsk and Benenitra. *The Planetary Science Journal*, 2, 244.
- Q. Xie, Z. Zhang*, **Z. Jin***, M. Santosh, L. Han, K. Wang, P. Zhao, H. He (2021), The high-grade Fe skarn deposit of Jinling, North China Craton: Insights into hydrothermal iron mineralization. *Ore Geology Reviews*, 138, 104395.
- **Z. Jin*** and M. Bose (2021), Hydration of Nebular Minerals through the Implantation–Diffusion Process. *The Astrophysical Journal*, 913, 116.
- **Z. Jin*** and M. Bose (2019), New clues to ancient water on Itokawa. *Science Advances*, eaav8106.

Full publication list available on [Google Scholar](#).

Students & Postdoc Supervision

Postdoc: Dr. Xiandi Zeng (2023–present)

Ph.D. Student: Pengfei Zhang (2022–2024), Jinwei Xiang (2025–2028)

Master Students: Zhirui Zhang (2022–2024), Jinwei Xiang (2022–2025), Luning Gu (2024–2026), Yongning Zhi (2025–)

Teaching history

Fall 2021 – 2024	Earth science (Undergraduate)
Spring 2022 –	Cosmochemistry (Master)
	Cosmochemistry and advanced topics (Ph.D)

Technical skills

- Operation on NanoSIMS 50L for isotopic analyses (e.g. H, Li, C, O, and S).
- Operation on Cameca SIMS-1280 HR for high precision isotopic analyses (e.g. U-Pb, O, B, and C).
- Operation on LA-ICPMS.
- Operation on Electronic Microprobe and Scanning Electron Microscope.
- Statistics of geochemical metrology.

- Preparation of samples for SIMS measurements (polishing, mounting, Au coating, optical microscope).